

Terraform RDS Module, Application and Web Files

File Structure

```
modules/  
  rds/  
    main.tf  
    variables.tf  
    outputs.tf  
  
application/  
  submit.php  
  
web/  
  index.html
```

modules/rds/main.tf

```
resource "aws_db_subnet_group" "main" {  
  name          = "three-tier-db-subnet-group"  
  subnet_ids    = var.private_db_subnet_ids  
  
  tags = {  
    Name = "three-tier-db-subnet-group"  
  }  
}  
  
resource "aws_db_instance" "main" {  
  identifier      = "three-tier-mysql"  
  engine          = "mysql"  
  engine_version  = "8.0"  
  instance_class  = var.instance_class  
  
  allocated_storage = 20  
  storage_type       = "gp3"  
  
  db_name          = var.db_name  
  username         = var.db_username  
  password         = var.db_password  
  
  db_subnet_group_name = aws_db_subnet_group.main.name  
  vpc_security_group_ids = [var.db_security_group_id]  
  
  publicly_accessible = false  
  skip_final_snapshot = true  
  
  tags = {  
    Name = "three-tier-mysql"  
    Tier = "Database"  
  }  
}
```

modules/rds/variables.tf

```
variable "vpc_id" {  
  type = string  
}  
  
variable "private_db_subnet_ids" {  
  type = list(string)  
}  
  
variable "db_name" {
```

```

    type = string
}

variable "db_username" {
    type = string
}

variable "db_password" {
    type      = string
    sensitive = true
}

variable "instance_class" {
    type = string
}

variable "db_security_group_id" {
    type = string
}

```

modules/rds/outputs.tf

```

output "rds_endpoint" {
    value = aws_db_instance.main.endpoint
}

output "rds_address" {
    value = aws_db_instance.main.address
}

output "rds_port" {
    value = aws_db_instance.main.port
}

```

application/submit.php

```

<?php

$host = getenv('DB_HOST');
$db   = getenv('DB_NAME');
$user = getenv('DB_USER');
$pass = getenv('DB_PASSWORD');

if (!$host || !$db || !$user || !$pass) {
    die("Database configuration is not available.");
}

$conn = new mysqli($host, $user, $pass, $db);

if ($conn->connect_error) {
    die("Database connection failed: " . $conn->connect_error);
}

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $name  = $_POST["name"] ?? "";
    $email = $_POST["email"] ?? "";

    $stmt = $conn->prepare(
        "INSERT INTO registrations (name, email) VALUES (?, ?)"
    );

    $stmt->bind_param("ss", $name, $email);

    if ($stmt->execute()) {

```

```

        echo "Registration successful.";
    } else {
        echo "Registration failed.";
    }

    $stmt->close();
}

$conn->close();
?>

```

web/index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>3-Tier AWS Application</title>
</head>

<body>

    <h1>3-Tier AWS Web Application</h1>

    <p>
        This application is deployed using Terraform on AWS.
    </p>

    <form action="/submit.php" method="POST">

        <label>Name:</label>
        <input type="text" name="name" required>

        <br><br>

        <label>Email:</label>
        <input type="email" name="email" required>

        <br><br>

        <button type="submit">Submit</button>

    </form>

</body>
</html>

```